

Prove that this integer must be odd.

Solution: It's easy to verify that from the numbers $1, 2, \dots, 2n$, there are n odd integers and n even integers.

At each step, we are removing two arbitrary integers j and K and introducing a new integer $|j-K|$.

$\therefore$ At each step, we are removing exactly one integer (Net effect).

After $2n-1$ steps of the process, there will be finally exactly one integer on the board (After $2n-1$ subtractions).

At each step, the following cases arise:

① j is even, K is even. By defn. 1, $\exists$ integers K_1, K_2 s.t. $j = 2K_1$ and $K = 2K_2$.

$$\therefore (j-K) = 2K_1 - 2K_2 = 2(K_1 - K_2). \text{ By Defn. 1, } (j-K) \text{ is even.}$$

$$\text{If } (j-K) > 0 \therefore |j-K| = j-K \text{ which is even}$$

$$\text{Else, } |j-K| = -(j-K) = -2(K_1 - K_2) = 2(K_2 - K_1). \text{ By Defn. 1, } |j-K| \text{ is even.}$$

② j is odd, K is even. By defn. 1, $\exists$ integers K_1, K_2 s.t. $j = 2K_1 + 1$ and $K = 2K_2$.

$$\therefore (j-K) = (2K_1 + 1) - 2K_2 = 2(K_1 - K_2) + 1. \text{ By Defn. 1, } (j-K) \text{ is odd}$$

$$\text{If } (j-K) > 0 \therefore |j-K| = j-K \text{ which is odd}$$

$$\text{Else, } |j-K| = -(j-K) = -2(K_1 - K_2) - 1 = 2(K_2 - K_1 - 1) + 1$$

$$\text{By Defn. 1, } |j-K| \text{ is odd.}$$

③ j is even, K is odd. By defn. 1, $\exists$ integers K_1, K_2 s.t. $j = 2K_1$ and $K = 2K_2 + 1$.

$$\therefore (j-K) = 2K_1 - (2K_2 + 1) = 2(K_1 - K_2 - 1) + 1. \text{ By Defn. 1, } (j-K) \text{ is odd}$$

$$\text{If } (j-K) > 0 \therefore |j-K| = j-K \text{ which is odd}$$

$$\text{Else, } |j-K| = -(j-K) = -2(K_1 - K_2 - 1) - 1 = 2(K_2 - K_1) + 2 - 1 = 2(K_2 - K_1) + 1$$

$$\text{By Defn. 1, } |j-K| \text{ is odd.}$$